

KALASIN ENGINEERING CO., LTD.

PROJECT SPRINT 18K TPA SPRAY DRYER / JET COOKER | REF: x2608003

STANDARD OPERATING PROCEDURE (SOP)

INSTRUMENT LOOP TESTING & FIELD COMMISSIONING METHODOLOGY

Conforming to IEC 62381, ISA-RP60.8, and NFPA 79 Standards

Document Field			Specification	Document Field			Specification
Document	Refer-		KAL-SOP-INS-0545	Issue Date:			07-October-2026
Revision:			Rev. 1.0 (Approved for Execution)	Target Facility:			Ingredion Kalasin Facility
Lead C&I	Contrac-		Kalasin Engineering Co., Ltd.	Client	Organiza-		Ingredion (Thailand) Co., Ltd.
				tion:			
Applicable	Enclo-		15 Junction Boxes (JB-401 to JB-618, IS-JBs)	Total Scope:			395 Equipment / 620 Channels
sures:							

1. PURPOSE & APPLICABLE STANDARDS

This Standard Operating Procedure (SOP) defines the mandatory technical requirements, safety precautions, test connection topologies, and acceptance criteria for performing instrument loop checks and pre-commissioning verification. All tests must strictly comply with:

- IEC 62381: Process Industry — Automation Systems — Activities during factory acceptance testing, site acceptance testing, and site integration.
- ISA-RP60.8: Hardware Guide for Control Centers & Instrument Loop Verification.
- IEC 60079-14: Explosive Atmospheres — Electrical installations design, selection, and erection (for Intrinsically Safe loops in IS-JB-603, 608, 612, 618).
- NFPA 79: Electrical Standard for Industrial Machinery.

2. PRE-COMMISSIONING COLD LOOP TEST METHODS

CRITICAL SAFETY PRECAUTION: SENSITIVE ELECTRONICS ISOLATION

High-voltage Megger insulation testing (500 VDC) **WILL PERMANENTLY DESTROY** solid-state instrument electronics, SMART positioners, and Allen-Bradley ControlLogix 1756 / 1794 FLEX I/O modules. Both cores (+ and -) **MUST BE FULLY DISCONNECTED** at the transmitter terminal head and at the CA1 marshalling rack before pressing the Megger test button.

2.1 Cable Insulation Resistance Testing (Megger 500 VDC)

Objective: Verify the dielectric integrity of cable insulation to prove zero leakage paths exist between conductors, shields, and ground.

- Test Instrument:** Calibrated 500 VDC Insulation Tester (e.g., Megger MIT410/2 or Fluke 1507). Calibration certificate must be valid within 12 months.
- Test Duration:** 60 seconds continuous test per measurement point.
- Acceptance Criterion:** ≥ 20 MΩ @ 500 VDC (Typical healthy new cabling reads > 100 MΩ).

Test Execution Matrix (4 Mandatory Test Points):

- Core-to-Core (A B):** Apply 500 VDC between Conductor A (+) and Conductor B (-). Minimum 20 MΩ.
- Core-to-Shield (A SH & B SH):** Apply 500 VDC between each signal conductor and the cable shield drain wire. Minimum 20 MΩ.
- Core-to-Earth (A PE & B PE):** Apply 500 VDC between each signal conductor and the plant Protective Earth (PE). Minimum 20 MΩ.
- Shield-to-Earth (SH PE):** Apply 500 VDC between the cable shield drain wire and Protective Earth. Minimum 20 MΩ (proves shield is floating at field/JB).

5. **Post-Test Discharge:** Allow the tester to automatically bleed off cable capacitive charge for at least 15 seconds before touching any conductor.

2.2 Loop Conductor Resistance Testing (Continuity Check)

Objective: Confirm complete end-to-end electrical path, verify that no high-resistance cold-solder or loose screw connections exist, and confirm correct polarity.

- **Test Instrument:** Digital Multimeter (Fluke 87V or Fluke 789 ProcessMeter) set to Low-Ohm resistance mode.
- **Acceptance Criterion:** $\leq 2.0 \Omega$ total round-trip loop resistance (typical for 0.75 mm^2 to 1.5 mm^2 cable runs up to 150 meters).

Step-by-Step Test Procedure:

1. **Lead Nulling:** Touch meter test leads together and engage the REL / ZERO function to null out test lead internal resistance (0.00Ω).
2. **Field Shorting:** At the field device terminal strip, attach an insulated temporary shorting bridge securely across Terminal 1 (+) and Terminal 2 (-).
3. **Resistance Measurement:** At the CA1 marshalling rack, measure resistance across the corresponding incoming trunk terminal pair. Record measured ohms.
4. **Open-Circuit Check:** Remove the field shorting bridge. The meter at CA1 must immediately transition to **O.L (Open Loop / Infinite Ω)**. If any finite resistance is observed, an unwanted parallel path or short circuit exists and must be repaired.
5. **Polarity Verification:** Apply a known +9.0 VDC battery across Core A (+) and Core B (-) at CA1. Measure DC voltage at the field device. Verify positive polarity on Core A.

2.3 Single-Point Earth Shielding (“CUTBACK & TAPE” Rule)

Objective: Prevent low-frequency 50 Hz AC ground hum, radio-frequency interference (RFI), and circulating eddy currents caused by differences in earth potential across the facility.

- **Acceptance Criteria:**
 - ▶ Shield-to-Ground resistance at Field Instrument: **Infinite (O.L / $> 20 \text{ M}\Omega$)** (Completely floating).
 - ▶ Shield-to-Ground resistance at Junction Box: **Infinite (O.L / $> 20 \text{ M}\Omega$)** (Isolated from metallic chassis).
 - ▶ Shield-to-IE Busbar at CA1 Marshalling: **$< 0.1 \Omega$** (Bonded cleanly).
 - ▶ Instrument Clean Earth (IE) to Ground Pit: **$< 1.0 \Omega$** .

Execution Rules:

- **At Field Device:** Strip the outer cable jacket. Cut the aluminum foil shield and bare tinned copper drain wire flush with the sheath base. Encase the cut end in a heat-shrink insulating sleeve and wrap securely with self-amalgamating tape. **The shield must never contact the metallic instrument enclosure.**
- **At Junction Box:** Connect the incoming branch shield drain and the outgoing trunk multi-pair shield drain through an **insulated feedthrough terminal (SH)**. Keep completely isolated from the mounting DIN rail and enclosure chassis.
- **At CA1 Marshalling Cabinet:** Land all trunk shield drain wires directly onto the dedicated, insulated copper **Instrument Earth (IE)** busbar. This represents the single, centralized earthing reference for the entire control loop.

2.4 Enclosure, Cable Gland & Tag Physical Inspection

Objective: Guarantee mechanical robustness, environmental sealing, and long-term operating reliability.

Inspection Item	Standard / Acceptance Spec	Verification Method
Ingress Protection (IP66/67)	Neoprene gasket intact, continuous, unpinched	Visual check; verify cover bolts torqued to 3.5 N·m
Cable Glands	Certified Ex-d / Ex-e nickel-plated brass or SS316	Hand tightness check; verify elastomeric seal compressed
Breather / Drain Plug	SS316 certified drain plug at lowest enclosure point	Verify downward orientation for condensate drainage
Stainless Steel Tag Plate	SS316 engraved plate (Tag, P&ID, Service)	Cross-reference with P&ID; verify secured with SS wire
Wire Terminations	Insulated bootlace ferrules on all stranded wires	Pull test (gentle tug); check torque (0.5–0.6 N·m)

Inspection Item	Standard / Acceptance Spec	Verification Method
Spare Cable Cores	Fitted with insulated end ferrules, labeled 'SPARE'	Neatly coiled in lower duct; landed on spare terminals

3. LIVE HOT LOOP & CALIBRATION METHODOLOGY

3.1 Analog Input (AI) 2-Pin Differential Pair Calibration

In the Allen-Bradley 1756-IF16 current input architecture, each analog channel utilizes **two physical pins**:

- **Pin A:** IN-x (Signal current input positive, landed on terminal strip (A))
- **Pin B:** i RTN-x (Signal current return negative, landed on terminal strip (B))

5-Point Simulated Signal Injection Table: Inject calibrated current using a Fluke 789 / Fluke 754 process calibrator directly across Pin A and Pin B at the field junction box:

Test Point	Simulated Input	PLC Raw Counts	SCADA Display Value	Max Error	Result
0% (Zero)	4.000 mA	0 counts	0.0 % (LRV)	± 0.20%	PASS
25% (Span 1/4)	8.000 mA	8192 counts	25.0 % EU	± 0.20%	PASS
50% (Mid)	12.000 mA	16384 counts	50.0 % EU	± 0.20%	PASS
75% (Span 3/4)	16.000 mA	24576 counts	75.0 % EU	± 0.20%	PASS
100% (Full Span)	20.000 mA	32767 counts	100.0 % (URV)	± 0.20%	PASS

3.2 Discrete Valve & Multi-Signal Equipment Testing

For actuated on-off valves (XV-40201A, HV, SOV) and flowmeters (FT-40201):

1. **Solenoid Actuation (DO):** Force DO bit in ControlLogix. Verify 24 VDC energizes the solenoid coil, pilot valve shifts cleanly, and valve strokes within specification (< 1.5 sec).
2. **Open / Close Limit Switches (DI):**
 - When valve reaches 100% full open: Proximity target hit, Dry Contact closes, PLC DI_1 bit changes 0 -> 1, SCADA icon turns steady **GREEN**.
 - When valve de-energizes to 0% full closed: Spring returns valve to fail-safe seat, PLC DI_2 bit changes 0 -> 1, SCADA icon turns steady **RED**.
3. **Flowmeter Aux Power & Pulse Counters:** Verify 24 VDC / 220 VAC auxiliary instrument power bus voltage. Inject 100 Hz pulse train into high-speed digital input (DI_1). Verify totalizer increments accurately on SCADA.

4. TRIPARTITE COMMISSIONING ACCEPTANCE SIGN-OFF

Upon completion of all cold and hot loop checks:

1. **Green Loop Test Tag:** Attach a weather-proof green inspection tag to the instrument cable gland marked with Date, Loop Pass Status, and Inspector Initials.
2. **Certificate Archiving:** Sign and date the individual equipment sheet inside the corresponding Junction Box Workbook (Loop_Test_Report_JB-xxx.xlsx).
3. **Tripartite Signatories:**
 - **Tested By:** Lead C&I Engineer, Kalasin Engineering Co., Ltd.
 - **Witnessed By:** QA/QC Electrical & Instrumentation Inspector, Ingredion (Thailand) Co., Ltd.
 - **Approved By:** Project Commissioning Manager, Joint Commissioning Team.